From Carb Counting to Diagnosis: Real World Patient Uses and Attitudes Toward Large Language Models in Diabetes Management

Poster Number: 250
Presentation Time: 05:45 PM - 07:00 PM

Abstract Keywords: Large Language Models (LLMs), Patient Engagement and Preferences, Chronic Care Management
Programmatic Theme: Consumer Health Informatics

Managing diabetes-related conditions is time-intensive and cognitively demanding for patients and caregivers, requiring ongoing glucose monitoring, dietary regulation, physical activity planning, and continuous lifestyle adaptation. With the emergence of large language models (LLMs), patients have increasingly turned to these tools for information, guidance, and support. However, there is limited empirical understanding of which diabetes-related medical tasks patients delegate to LLMs and what their experiences are. To address this gap, we combined qualitative thematic analysis with LLM-assisted analysis to examine patient attitudes and real-world use cases in using LLMs for diabetes-related tasks. Our analysis identified diverse application areas, ranging from clinical interpretation to nutrition and diet support, and disease management amongst others. LLMs functioned not only as information sources, but as interpretive, analytical, decision-support, emotional, and logistical aids supporting patients' self-management. Last, we discuss implications for integrating LLMs into patients' self-management support ecosystems and identify areas that require support and safeguards.

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